



On the edge-connectivity of the square of a graph

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ARTICLE INFO

Article history:

Received 28 August 2024

Received in revised form 31 December 2024

Accepted 17 January 2025

Available online 31 January 2025

Keywords:

Edge-connectivity

Maximally edge-connected graph

Square

Connectivity

ABSTRACT

Let G be a connected graph. The edge-connectivity of G , denoted by $\lambda(G)$, is the minimum number of edges whose removal renders G disconnected. Let $\delta(G)$ be the minimum degree of G . It is well-known that $\lambda(G) \leq \delta(G)$, and graphs for which equality holds are said to be maximally edge-connected. The square G^2 of G is the graph with the same vertex set as G , in which two vertices are adjacent if their distance is not more than 2.

In this paper we present results on the edge-connectivity of the square of a graph. We show that if the minimum degree of a connected graph G of order n is at least $\lfloor \frac{n+2}{4} \rfloor$, then G^2 is maximally edge-connected, and this result is best possible. We also give lower bounds on $\lambda(G^2)$ for the case that G^2 is not maximally edge-connected: We prove that $\lambda(G^2) \geq \kappa(G)^2 + \kappa(G)$, where $\kappa(G)$ denotes the connectivity of G , i.e., the minimum number of vertices whose removal renders G disconnected, and this bound is sharp. We further prove that $\lambda(G^2) \geq \frac{1}{2}\lambda(G)^{3/2} - \frac{1}{2}\lambda(G)$, and we construct an infinite family of graphs to show that the exponent $3/2$ of $\lambda(G)$ in this bound is best possible.

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1. Introduction

The *edge-connectivity* $\lambda(G)$ of a graph G is defined as the smallest number of edges whose removal renders G disconnected. It is one of the oldest and best studied graph invariants. A classical result by Whitney [17] states that the edge-connectivity of a graph cannot exceed its *minimum degree*, defined as the smallest of the degrees of the vertices of G . Graphs whose edge-connectivity equals the minimum degree are called *maximally edge-connected* graphs. Much research on edge-connectivity has focused on finding sufficient conditions for a graph to be maximally edge-connected, see for example the survey by Hellwig and Volkmann [12]. Of particular relevance for us is the following classical result by Chartrand [4].

Theorem 1. *Let G be a graph of order n and minimum degree $\delta(G)$. If*

$$\delta(G) \geq \frac{n-1}{2},$$

then $\lambda(G) = \delta(G)$.

Also results on the edge-connectivity of graphs derived from other graphs can be found in the literature, for example line graphs (see [3,5,10,13]), products of graphs (see [15]) and complements of graphs (see, for example, [1,2,11]).

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The *square* G^2 of a graph G is the graph on the same vertex set as G , in which two vertices are adjacent if their distance in G is not more than two. Somewhat surprisingly, the edge-connectivity of the square of a graph appears not to have been studied. In this paper we aim to fill this gap in the literature.

This paper is organised as follows. The terminology in this paper is defined in Section 2. In Section 3 we give a minimum degree condition on G guaranteeing that G^2 is maximally edge-connected; we prove that if a connected graph G of order n satisfies the condition $\delta(G) \geq \lfloor \frac{n+2}{4} \rfloor$, then $\lambda(G^2) = \delta(G^2)$. We also construct a family of graphs to show that this condition is best possible. In the following two sections we relate connectivity-parameters of G to the edge-connectivity of G^2 . In Section 4 we give a lower bound on $\lambda(G^2)$ in terms of $\kappa(G)$, the connectivity of G . We show that $\lambda(G^2) \geq \kappa(G)^2 + \kappa(G)$, unless G^2 is maximally edge-connected, and this bound is sharp. In Section 5 we give a lower bound on $\lambda(G^2)$ in terms of $\lambda(G)$. We prove that $\lambda(G^2) \geq \frac{1}{2}\lambda(G)^{3/2} - \frac{1}{2}\lambda(G)^{1/2}$, unless G is maximally edge-connected. We construct an infinite family of graphs showing that the exponent $3/2$ of $\lambda(G)$ is best possible.

2. Terminology and notation

The notation we use is as follows. G denotes a connected, finite graph with vertex set $V(G)$ and edge set $E(G)$. The *order* of G , i.e., the number of vertices, is denoted by $n(G)$. An edge joining a vertex u to a vertex v is denoted by uv . If v is a vertex of G then we write $N_G(v)$ for the *neighbourhood* of v , i.e., the set of all vertices of G adjacent to v , and $N_G[v]$ for the *closed neighbourhood* of v , i.e., the set $N_G(v) \cup \{v\}$. The *degree* $\deg_G(v)$ of v is defined as $|N_G(v)|$, and the *minimum degree* of G , denoted by $\delta(G)$, is $\min_{w \in V(G)} \deg_G(w)$. If there is no danger of confusion, then we often drop the argument or subscript G .

The *distance* $d_G(u, v)$ between u and v is the minimum length of a (u, v) -path. An *edge-cut* of G is a set of edges whose removal disconnects the graph, and a *minimum edge-cut* is an edge-cut of cardinality $\lambda(G)$.

If A and B are disjoint subsets of the vertex set of G , then $E_G(A, B)$ denotes the set of all edges of G joining a vertex of A to a vertex of B . The subgraph of G *induced* by A , denoted by $G[A]$, is the graph with vertex set A , in which two vertices are adjacent if and only if they are adjacent in G .

By K_n we mean the complete graph on n vertices. The *sequential sum* of disjoint graphs $G_1, G_2, \dots, G_k$, denoted by $G_1 + G_2 + \dots + G_k$, is the graph obtained from their union by joining every vertex in G_i to every vertex in G_{i+1} for $i = 1, 2, \dots, k-1$.

In the proofs below we use the following notation. Given a connected graph G , we denote the graph G^2 by H . The set $S \subseteq E(H)$ is always a minimum edge-cut of H . The two components of $H - S$ are denoted by H_1 and H_2 , and their vertex sets by V_1 and V_2 , respectively. A and B are the sets of vertices in H_1 and H_2 , respectively, incident with edges in S . A vertex of H_1 or H_2 is an *interior vertex* if it is not in A or B , respectively. We let $S' = S \cap E(G)$. Then $G - S'$ is disconnected since it is a spanning subgraph of $H - S$. Let G_1 and G_2 be the subgraphs of G induced by V_1 and V_2 . Then G_1 and G_2 are unions of components of $G - S'$. Let A' and B' be the set of vertices of G_1 and G_2 , respectively, incident with edges in S' . Note that $A' \subseteq A$ and $B' \subseteq B$.

3. A sufficient condition for G^2 to be maximally edge-connected

Theorem 1 shows that every graph G of order n with $\delta(G) \geq \frac{n-1}{2}$ is maximally edge-connected. In this section we show that this condition can be relaxed considerably to guarantee that G^2 is maximally edge-connected. In our proof we make use of the following well-known lemma.

Lemma 1 (Hamidoune [9]). *Let G be a connected graph. If $S \subseteq E(G)$ is an edge-cut with $|S| < \delta(G)$, then every component of $G - S$ has an interior vertex.*

Theorem 2. *Let G be a connected graph of order n . If*

$$\delta(G) \geq \left\lfloor \frac{n+2}{4} \right\rfloor,$$

then G^2 is maximally edge-connected.

Proof. Suppose that H is not maximally edge-connected. Let $H, S, H_1, H_2, S', A', B', V_1, V_2$, be as defined above.

Claim 1. $|S| \leq 2\delta(G) - 2$.

Suppose to the contrary that $|S| \geq 2\delta(G) - 1$. Applying the hypothesis of the theorem, we obtain that

$$|S| \geq 2\delta(G) - 1 \geq 2\left\lfloor \frac{n+2}{4} \right\rfloor - 1 \geq 2\frac{n-1}{4} - 1 = \frac{n-3}{2}.$$

Since H is not maximally edge-connected, we thus have $\delta(H) \geq |S| + 1 \geq \frac{n-1}{2}$. It follows by **Theorem 1** that H is maximally edge-connected. This contradiction proves **Claim 1**.

Claim 2. For all $u \in A' \cup B'$ we have $|N_G[u] \cap V_1| = 1$ or $|N_G[u] \cap V_2| = 1$.

Clearly $|N_G[u] \cap V_i| \geq 1$ for all $u \in A' \cup B'$ and $i \in \{1, 2\}$. Suppose, to the contrary, that there exists $u \in A' \cup B'$ with $|N_G[u] \cap V_1| \geq 2$ and $|N_G[u] \cap V_2| \geq 2$. Without loss of generality, we may assume that $u \in A'$. Since $N_G[u]$ induces a complete graph of order $\deg_G(u) + 1$ in H , we have that

$$\begin{aligned} |S \cap E(H[N_G[u]])| &\geq |N_G[u] \cap V_1| |N_G[u] \cap V_2| \\ &= |N_G[u] \cap V_1| (\deg_G(u) + 1 - |N_G[u] \cap V_1|) \\ &\geq 2(\deg_G(u) - 1) \\ &\geq 2\delta(G) - 2. \end{aligned} \quad (1)$$

From (1) and Claim 1 we obtain that $|S \cap E(H[N_G[u]])| = |S| = 2\delta(G) - 2$, and so

$$S \subseteq E(H[N_G[u]]). \quad (2)$$

This implies in particular that $B' \subseteq N_G(u)$. By Lemma 1, H_2 contains an internal vertex v_2 . Let Q be a shortest path in G from B' to v_2 . Let $w \in B'$ be the initial vertex of Q and x its successor. Then $ux \in E(H)$. Since clearly, $x \in V_2$, we have $ux \in S$. But $x \notin N_G[u]$, and so $ux \notin E(H[N_G[u]])$. This contradiction to (2) proves Claim 2.

Claim 3. There exist $u \in A'$, $v \in B'$ with $|N_G[u] \cap V_2| = |N_G[v] \cap V_1| = 1$.

By Lemma 1 there exists an internal vertex v_1 of H_1 , i.e., a vertex not in A and thus not in A' . Let P be a shortest path in G from v_1 to A' , let u be the terminal vertex of P and y its predecessor. Then $y \in V_1$, and so $\{u, y\} \subseteq N_G[u] \cap V_1$. By Claim 1 we have $|N_G[u] \cap V_2| = 1$, as desired. Similarly we show the existence of a vertex $v \in B'$ with $|N_G[v] \cap V_1| = 1$. Claim 3 follows.

We are now ready to complete the proof.

Let u and v be as in Claim 3, and let u^* and v^* be the unique neighbours of u and v in V_2 and V_1 , respectively, in G . Let $E_u \subseteq E(H)$ be the set of all edges of the form xu^* , where $x \in N_G[u] \cap V_1$, and similarly let $E_v \subseteq E(H)$ be the set of all edges of the form xv^* , where $x \in N_G[v] \cap V_2$. Clearly,

$$E_u \cup E_v \subseteq S.$$

Clearly, $|E_u| = \deg_G(u) \geq \delta(G)$, and similarly $|E_v| \geq \delta(G)$. Since all edges in E_u are incident with u^* , and all edges in E_v are incident with v^* , we conclude that $E_u \cap E_v$ is either empty or contains the edge u^*v^* but no further edge. Hence

$$|S| \geq |E_u| + |E_v| - |E_u \cap E_v| \geq 2\delta(G) - 1.$$

This contradiction to Claim 2 concludes the proof of the theorem. $\square$

The condition $\delta(G) \geq \lfloor \frac{n+2}{4} \rfloor$ in Theorem 2 is best possible for all n with $n \geq 10$. For given n let G_n be the sequential sum

$$K_{n/4} + K_{(n-4)/4} + K_1 + K_1 + K_{(n-8)/4} + K_{(n+4)/4} \quad \text{if } n \equiv 0 \pmod{4},$$

$$K_{(n+3)/4} + K_{(n-5)/4} + K_1 + K_1 + K_{(n-9)/4} + K_{(n+3)/4} \quad \text{if } n \equiv 1 \pmod{4},$$

$$K_{(n+2)/4} + K_{(n-6)/4} + K_1 + K_1 + K_{(n-6)/4} + K_{(n+2)/4} \quad \text{if } n \equiv 2 \pmod{4},$$

$$K_{(n+1)/4} + K_{(n-7)/4} + K_1 + K_1 + K_{(n-7)/4} + K_{(n+5)/4} \quad \text{if } n \equiv 3 \pmod{4}.$$

It is easy to verify that $\delta(G_n) = \lfloor \frac{n+2}{4} \rfloor - 1$, but $\lambda(G_n^2) = \delta(G_n^2) - 1$.

4. A lower bound on $\lambda(G^2)$ in terms of $\kappa(G)$

In this section we explore the relationship between the connectivity of a graph and the edge-connectivity of its square. For graphs whose square is not maximally edge-connected, we give a lower bound on the edge-connectivity of G^2 in terms of the minimum degree and the connectivity of G . As a corollary we obtain a sharp lower bound on the edge-connectivity of G^2 in terms of the connectivity of G only.

Theorem 3. Let G be a connected graph with $\delta(G) \geq 2$. Then G^2 is maximally edge-connected, or

$$\lambda(G^2) \geq \kappa(G)(\delta(G) + 1).$$

Proof. Let H, S, H_1, H_2 and S' be as defined above. We may assume that H is not maximally edge-connected, otherwise the theorem holds. By Lemma 1 there exist interior vertices v_1 of H_1 and v_2 of H_2 .

Let $\kappa := \kappa(G)$ and $\delta := \delta(G)$. By Menger's Theorem, there exist κ internally vertex-disjoint (v_1, v_2) -paths $P_1, P_2, \dots, P_\kappa$ in G . We may assume that each P_i is an induced path in G . Each P_i uses at least one edge of S' . Let $a_i b_i$ be the last edge of

P_i in S' .

We now show that

$$|S| \geq \kappa + \sum_{i=1}^{\kappa} \deg_G(a_i). \quad (3)$$

Let $M_1 := \{a_1, a_2, \dots, a_\kappa\}$. Consider the edges of G incident with a vertex in M_1 . First let e be an edge incident with exactly one vertex in M_1 , so $e = a_i x$ for some $i \in \{1, 2, \dots, \kappa\}$ and $x \in V(G) - M_1$. From e we obtain an edge $e_S \in S$ as follows. If $x \in V_2$, then let $e_S = e$. If $x \in V_1 - M_1$, then let $e_S = b_i x$. Now let e be an edge incident with two vertices in M_1 , so $e = a_i a_j$ for some $i, j \in \{1, 2, \dots, \kappa\}$. We obtain two edges from e , the edges $a_i y_j$ and $a_j y_i$, where y_j is the first vertex of P_j after a_j that is not in $N_G(a_i)$, and y_i is the first vertex of P_i after a_i that is not in $N_G(a_j)$. Clearly, $y_i, y_j \in V_2$, so $a_i y_j, a_j y_i \in S$. Note that $y_i, y_j \neq v_2$ since otherwise v_2 would not be an internal vertex of H , and so $y_i \neq y_j$.

Let E_1 be the set of edges of S thus obtained from the edges incident with vertices of M_1 . Then each edge in E_1 is either (i) of the form $x b_i$, where $x \in V_1 - M_1$, in which case it is obtained from the edge $a_i x$, or (ii) of the form $a_i x$, where $x \in V_2$ and $a_i x$ is an edge of G , in which case it is obtained from $a_i x$, or (iii) of the form $a_i x$, where $x \in V_2 \cap V(P_j)$ and $a_i x$ is not an edge of G . It follows that we do not obtain the same edge from two distinct edges of G incident with a vertex in M_1 . Therefore,

$$|E_1| = \sum_{i=1}^{\kappa} \deg_G(a_i).$$

For $i = 1, 2, \dots, \kappa$, let w_i be the vertex following b_i on P_i . Then $a_i w_i \in S$, but it follows from the above that $a_i w_i \notin E_1$, so S contains at least κ edges in addition to the edges in E_1 . Hence (3) follows.

Since each vertex of M_1 has degree at least δ in G , (3) yields that

$$|S| \geq |E_1| + \kappa \geq \kappa + \kappa \delta. \quad (4)$$

Since $|S| = \lambda(G^2)$, the theorem follows. $\square$

We do not know if the bound in Theorem 3 is sharp. However, using the inequality $\delta(G) \geq \kappa(G)$, we obtain the following sharp lower bound on the edge-connectivity of G^2 in terms of the connectivity of G as a corollary to Theorem 3.

Corollary 1. *Let G be a κ -connected graph with $\kappa \geq 2$. Then G^2 is maximally edge-connected or*

$$\lambda(G^2) \geq \kappa(\kappa + 1).$$

If κ is even, then the bound is sharp.

Proof. The bound in Corollary 1 follows from Theorem 3 in conjunction with the inequality $\kappa(G) \leq \delta(G)$.

It remains to prove the sharpness of the bound. For even $\kappa \in \mathbb{N}$ consider the graph $K_\kappa - M$, where K_κ is a complete graph with vertex set $\{u_1, u_2, \dots, u_\kappa\}$, and M is the perfect matching $\{u_1 u_2, u_3 u_4, \dots, u_{\kappa-1} u_\kappa\}$ of K_κ . For $i = 1, 2, 3, 4$ let G_i be a copy of $K_\kappa - M$, and denote the vertex of G_i corresponding to a vertex u_j of $K_\kappa - M$ by $u_j^{(i)}$. Let G_0 and G_5 be disjoint copies of the complete graph $K_{2\kappa^2}$. Then we obtain the graph $G(\kappa)$ from joining every vertex of G_0 to every vertex of G_1 , and every vertex of G_4 to every vertex of G_5 by an edge, and by adding the edges $v_i^{(1)} v_i^{(2)}, v_i^{(2)} v_i^{(3)}$ and $v_i^{(3)} v_i^{(4)}$ for $i = 1, 2, \dots, \kappa$. The graph $G(\kappa)$ for $\kappa = 4$ is sketched in Fig. 1.

It is easy to verify that $G(\kappa)$ is κ -connected. Denote the vertex set of G_i by V^i . Clearly, removing the edges in $E_{G^2}(V^0 \cup V^1 \cup V^2, V^3 \cup V^4 \cup V^5)$ disconnects the graph G^2 . Now $|E_{G^2}(V^2, V^3)| = \kappa(\kappa - 1)$, $|E_{G^2}(V^1, V^3)| = |E_{G^2}(V^2, V^4)| = \kappa$, and so

$$\begin{aligned} \lambda(G^2) &\leq |E_{G^2}(V^0 \cup V^1 \cup V^2, V^3 \cup V^4 \cup V^5)| \\ &= |E_{G^2}(V^2, V^3)| + |E_{G^2}(V^1, V^3)| + |E_{G^2}(V^2, V^4)| \\ &= \kappa(\kappa + 1). \end{aligned}$$

Since every vertex of G^2 is adjacent to all vertices of V^0 or all vertices of V^5 , except possibly itself, we have $\delta(G^2) \geq 2\kappa^2 > \lambda(G^2)$, so G^2 is not maximally edge-connected. Hence Corollary 1 yields that $\lambda(G^2) \geq \kappa(\kappa + 1)$, so G^2 attains equality in Corollary 1. $\square$

We note that for odd κ , there exist graphs for which the edge-connectivity of the square differs from the bound in Corollary 1 by 1. To see that consider, for odd $\kappa \in \mathbb{N}$, the graph $K_\kappa - M_0$, where K_κ is a complete graph with vertex set $\{u_1, u_2, \dots, u_\kappa\}$, and M is the minimum edge-cover $\{u_1 u_2, u_3 u_4, \dots, u_{\kappa-2} u_{\kappa-1}, u_{\kappa-1} u_\kappa\}$ of K_κ . For $i = 1, 2, 3, 4$, let G_i be a copy of $K_\kappa - M$, and denote the vertex of G_i corresponding to a vertex u_j of $K_\kappa - M$ by $u_j^{(i)}$. Let G_0 and G_5 be disjoint copies of the complete graph $K_{2\kappa^2}$. Then we obtain the graph $G(\kappa)$ from joining every vertex of G_0 to every vertex of G_1 , and every vertex of G_4 to every vertex of G_5 by an edge, and by adding the edges $v_i^{(1)} v_i^{(2)}, v_i^{(2)} v_i^{(3)}$ and $v_i^{(3)} v_i^{(4)}$ for $i = 1, 2, \dots, \kappa$. Finally we add the edges $u_{\kappa-1}^{(2)} u_\kappa^{(1)}$ and $u_{\kappa-1}^{(3)} u_\kappa^{(4)}$. The graph $G(\kappa)$ for $\kappa = 5$ is sketched in Fig. 2. It is easy to verify that $G(\kappa)$ is κ -connected. Similar considerations as for the case κ even show that

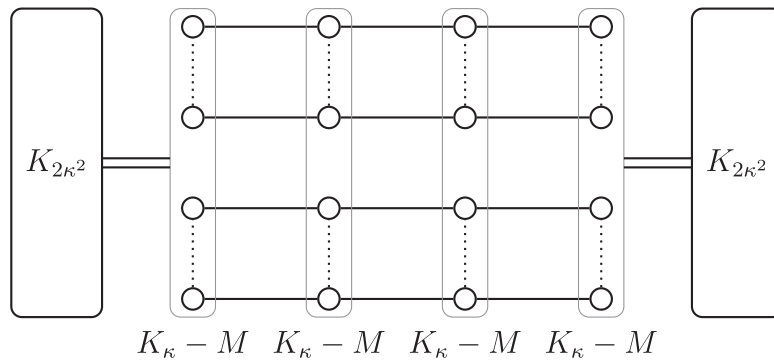


Fig. 1. The graph $G(\kappa)$ for $\kappa = 4$. Dotted lines indicate edges not present.

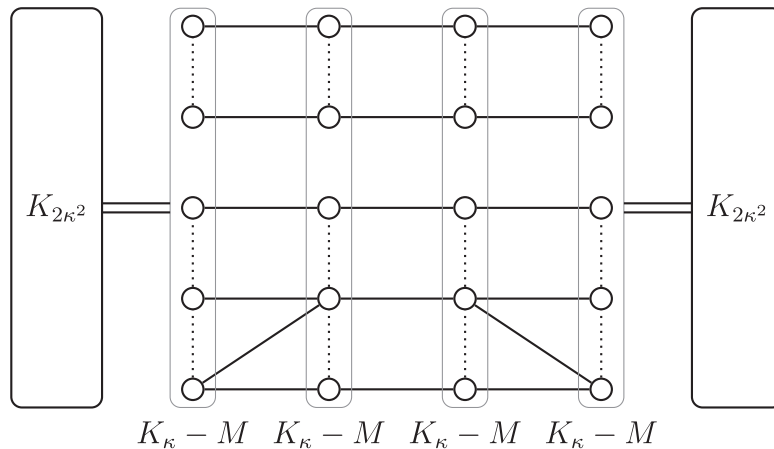


Fig. 2. The graph $G(\kappa)$ for $\kappa = 5$. Dotted lines indicate edges not present.

$$\lambda(G(\kappa)^2) \leq \kappa(\kappa + 1) + 1.$$

Corollary 2. Let G be a connected graph with minimum degree δ . If G is not complete, then

$$\lambda(G^2) \geq \delta + 1,$$

and this bound is sharp.

Proof. If G is not complete, then clearly $\delta(G^2) \geq \delta + 1$. Thus, if G^2 is maximally edge-connected, then the statement of the corollary holds. If G^2 is not maximally edge-connected, then $\kappa(G) \geq 1$, and the corollary follows from Theorem 3.

To see that the bound is sharp consider for given $\delta \in \mathbb{N}$ the graph $G = K_1 + K_\delta + K_1 + K_\delta$. Clearly $\delta(G) = \delta$ and $\lambda(G^2) = \delta(G^2) = \delta + 1$. $\square$

5. A lower bound on $\lambda(G^2)$ in terms of $\lambda(G)$

In this section we explore the relationship between the edge-connectivity of G and the edge-connectivity of G^2 . We show that $\lambda(G^2)$ is bounded from below by a term $O(\lambda(G)^{3/2})$, unless G^2 is maximally edge-connected. We also construct an infinite family of graphs showing that the exponent $3/2$ is best possible.

Theorem 4. Let G be a λ -edge-connected graph. Then G^2 is maximally edge-connected or

$$\lambda(G^2) \geq \frac{1}{2}\lambda^{3/2} - \frac{1}{2}\lambda.$$

Proof. Let H, S, A, B, S', A', B' be as defined above. For $v \in A' \cup B'$ we also define $\text{inc}(v, S')$ to be the number of edges in S' incident with v .

We may assume that H is not maximally edge-connected, otherwise the theorem holds.

CASE 1: There exists a vertex $v \in A' \cup B'$ with $\sqrt{\lambda} \leq \text{inc}(v, S') \leq \lambda - \lambda^{1/2}$.

We may assume that $v \in A'$. If $u \in N_G(v) \cap V_1$ and $w \in N_G(v) \cap V_2$, then the edge uw of H is in S . Hence

$$\begin{aligned} |S| &\geq |N_G(v) \cap V_1| \cdot |N_G(v) \cap V_2| \\ &= (\deg_G(v) - \text{inc}(v, S')) \text{inc}(v, S') \\ &\geq (\lambda - \text{inc}(v, S')) \text{inc}(v, S'). \end{aligned}$$

For a real x with $\lambda \leq x \leq \lambda - \lambda^{1/2}$, the function $(\lambda - x)x$ is minimised if $x = \sqrt{\lambda}$ or $x = \lambda - \lambda^{1/2}$. Substituting either of these values yields $\lambda(H) = |S| \geq \lambda^{3/2} - \lambda$, as desired.

CASE 2: For every $v \in A' \cup B'$ we have $\text{inc}(v, S') < \sqrt{\lambda}$ or $\text{inc}(v, S') > \lambda - \sqrt{\lambda}$.

We denote by A^+ and A^- the set of all $v \in A'$ with $\text{inc}(v, S') > \lambda - \sqrt{\lambda}$ and $\text{inc}(v, S') < \sqrt{\lambda}$, respectively. The sets B^+ and B^- are defined analogously. Then $A^+ \cup A^- = A'$, $B^+ \cup B^- = B'$.

Define the sets W_1 and W_2 by $W_1 = (V_1 - A^+) \cup B^+$ and $W_2 = (V_2 - B^+) \cup A^+$ and consider the edge-cut $T := E_G(W_1, W_2)$ of G . Then

$$T = E_G(A^+, B^+) \cup E_G(A^-, B^-) \cup E_G(A^+, V_1 - A^+) \cup E_G(B^+, V_2 - B^+). \quad (5)$$

Since G is λ -edge-connected, we have $|T| \geq \lambda$, and so at least one of the terms $|E_G(A^+, B^+)|$, $|E_G(A^-, B^-)|$, $|E_G(A^+, V_1 - A^+)|$ and $|E_G(B^+, V_2 - B^+)|$ is greater or equal to $\frac{1}{4}\lambda$. We consider the following (not mutually exclusive) subcases.

CASE 2A: $|E_G(A^+, B^+)| \geq \frac{1}{4}\lambda$.

Since $|E_G(A^+, B^+)| \leq |A^+| \cdot |B^+|$, we have $|A^+| \cdot |B^+| \geq \frac{1}{4}\lambda$ and so $|A^+| + |B^+| \geq \sqrt{\lambda}$. We may assume that $|A^+| \geq |B^+|$, and thus $|A^+| \geq \frac{1}{2}\sqrt{\lambda}$. Since each vertex of A^+ is incident with at least $\lambda - \sqrt{\lambda}$ edges of S' , we have

$$|S'| \geq \sum_{v \in A^+} \text{inc}(v, S') \geq \frac{1}{2}\sqrt{\lambda}(\lambda - \sqrt{\lambda}) = \frac{1}{2}\lambda^{3/2} - \frac{1}{2}\lambda. \quad (6)$$

Since $\lambda(H) = |S| \geq |S'|$, the theorem holds in this case.

CASE 2B: $|E_G(A^-, B^-)| \geq \frac{1}{4}\lambda$.

Let A^* (B^*) be the set of vertices in A^- that are adjacent in G to a vertex in B^- (A^-). The defining condition of this case yields that $|E_G(A^*, B^*)| \geq \frac{1}{4}\lambda$. We may assume that $|A^*| \geq |B^*|$, and the same arguments as in Case 2 A show that $|A^*| \geq \frac{1}{2}\sqrt{\lambda}$. Each vertex $v \in A^*$ has a neighbour w in B^- , and it follows from the definition of B^- that w has at least $\lambda - \sqrt{\lambda}$ neighbours in V_2 , and v is joined to each of these by an edge in H , which is in S . Hence

$$|S| \geq \sum_{v \in A^*} \text{inc}(v, S) \geq \frac{1}{2}\sqrt{\lambda}(\lambda - \sqrt{\lambda}) = \frac{1}{2}\lambda^{3/2} - \frac{1}{2}\lambda,$$

and the theorem holds also in this case.

CASE 2C: $|E_G(A^+, V_1 - A^+)| \geq \frac{1}{4}\lambda$ or $|E_G(B^+, V_2 - B^+)| \geq \frac{1}{4}\lambda$.

We may assume that $|E_G(A^+, V_1 - A^+)| \geq \frac{1}{4}\lambda$. Moreover, we may assume that $|A^+| < \frac{1}{2}\sqrt{\lambda}$ since otherwise, if $|A^+| \geq \frac{1}{2}\sqrt{\lambda}$, the same reasoning as in (6) shows that the theorem holds.

Now let $v_0 \in A^+$ be a vertex with the most neighbours in $V_1 - A^+$. Then

$$|N_G(v_0, V_1 - A^+)| \geq \frac{|E_G(A^+, V_1 - A^+)|}{|A^+|} \geq \frac{\frac{1}{4}\lambda}{\frac{1}{2}\sqrt{\lambda}} = \frac{1}{2}\sqrt{\lambda}.$$

Since every vertex in $N_G(v_0) \cap V_1$ is joined to every vertex in $N_G(v_0) \cap V_2$, by an edge in S , we have

$$|S| \geq |N_G(v_0, V_1 - A^+)| \cdot \text{inc}(v_0, S') \geq \frac{1}{2}\sqrt{\lambda} \cdot (\lambda - \sqrt{\lambda}) = \frac{1}{2}\lambda^{3/2} - \frac{1}{2}\lambda.$$

In all cases, the theorem holds. $\square$

The exponent $3/2$ of λ in the bound in Theorem 4 is best possible. To see this consider for given square integer $\lambda \geq 4$ the graph G_λ , defined by

$$G_\lambda = K_{\lambda^2} + K_{\lambda-\sqrt{\lambda}} + K_{\sqrt{\lambda}} + K_{\sqrt{\lambda}} + K_{\lambda-\sqrt{\lambda}} + K_{\lambda^2}.$$

It is easy to verify that G_λ is λ -edge-connected, that $\delta(G_\lambda^2) = \lambda^2 + \lambda - 1$, and that $\lambda(G_\lambda^2) = 2\lambda^{3/2} - \lambda$.

6. Conclusion and open problems

In Section 3, we presented a degree condition on a graph G that guarantees that G^2 is maximally edge-connected. Our condition in Theorem 2 is a relaxation of the condition $\delta(G) \geq \frac{n-1}{2}$ in Theorem 1. It is natural to ask if other sufficient conditions for maximally edge-connected graphs can be relaxed in similar ways to guarantee that the square of a graph is maximally edge-connected. Possible candidates are minimum degree conditions for graphs not containing certain subgraphs (see, for example, [6]), degree sequence conditions (see, for example, [8]), conditions in terms of the inverse degree, i.e., the sum of the reciprocals of the vertex degrees, (see [7]), conditions in terms of size [16], or conditions in terms of girth and diameter (see, for example [14]).

It is natural to expect that the lower bounds on the edge-connectivity of G^2 in terms of the connectivity of G in Section 4 and in terms of the edge-connectivity of G in Section 5 can be generalised to higher powers of graphs. For $k \in \mathbb{N}$, the k th power of G is the graph on the same vertex set as G , where two vertices are adjacent if their distance is not more than k . It would also be interesting to determine the smallest minimum degree of a graph G of given order that guarantees that its k th power is maximally edge-connected. We pose these questions as open problems.

Data availability

No data was used for the research described in the article.

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